

EE204 Semiconductor Devices

HW 6 (BJT)

1. A Si p⁺-n-p transistor has impurity concentrations of 5×10^{18} , 2×10^{17} and 10^{16} cm^{-3} in the emitter, base, and collector, respectively. The base width W_B is $1 \text{ } \mu\text{m}$ and the device cross-sectional area is 2 mm^2 . When the emitter-base junction is forward-biased to 0.5 V and the base-collector junction is reverse-biased to 5 V , calculate (a) the neutral base width and (b) the minority carrier concentration at $x=0$ in the neutral base.

(Ans. $0.904 \text{ } \mu\text{m}$ and $1.13 \times 10^{11} \text{ cm}^{-3}$)

2. For the transistor in Problem 1, the diffusion constants of minority carriers in the emitter, base, and collector are 52 , 40 , and $115 \text{ cm}^2/\text{s}$, respectively; and the corresponding lifetimes are 10^{-8} , 10^{-7} , and 10^{-6} s . Find the current component I_{Ep} , I_{Cp} , I_{En} , I_{Cn} , and I_{BB} .

(Ans. $1.596 \times 10^{-5} \text{ A}$, $1.596 \times 10^{-5} \text{ A}$, $1.04 \times 10^{-7} \text{ A}$, $3.196 \times 10^{-14} \text{ A}$, 0)

3. Using the results obtained from problem 1 and 2, (a) find the terminal currents I_E , I_C , and I_B , of the transistor; (b) calculate emitter efficiency, base transport factor, common-base current gain, and common-emitter current gain. (c) Comment on how the emitter efficiency and base transport can be improved.

(Ans.(a) $1.606 \times 10^{-5} \text{ A}$, $1.596 \times 10^{-5} \text{ A}$, $1.041 \times 10^{-7} \text{ A}$, (b) 0.9938 , 1 , 0.9938 , 160.3)

4. A Si transistor has D of $10 \text{ cm}^2/\text{s}$ and W of $0.5 \text{ } \mu\text{m}$. Find the cutoff frequencies for the transistor with a common base current gain of 0.998 . Neglect the emitter and collector delays. To design a bipolar transistor with 5 GHz cutoff frequency, what the neutral base width W will be?

(Ans:(a) $f = 1.27 \text{ GHz}$ and $f_{\beta} = 2.55 \text{ MHz}$;(b) $0.252 \text{ } \mu\text{m}$)